

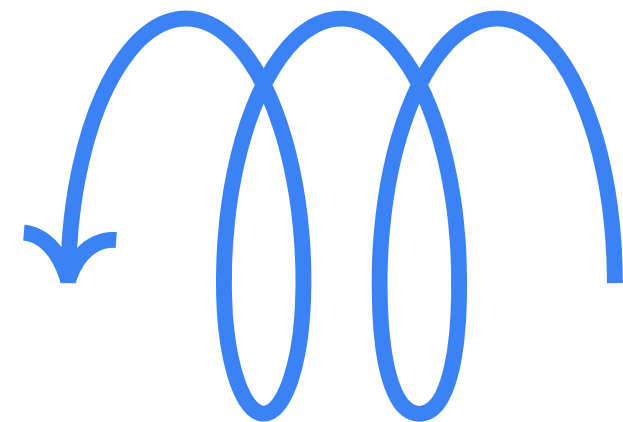
MULTI-INDUSTRY COUNTERFEIT DETECTION

Developing a single GNN-based platform that uses existing transaction data to flag counterfeit injection points across all industries.

By Team Little Boy



LIMITATIONS, RISKS, AND GAPS IN THE EXISTING SOLUTIONS



1

Label Scarcity

Confirmed counterfeit cases are rare, costly, and underreported, creating noisy labels and extreme class imbalance that existing GNN models struggle to handle.

3

Temporal Drift & Evolving Graphs

Existing models struggle with changing distributors, seasonal demand, and shifting routes, causing performance degradation and false positives during legitimate demand surges.

2

Siloed Data

Existing counterfeit detection systems struggle to integrate invoice, logistics, warehouse, distributor, and retailer data, limiting end-to-end supply chain visibility.

4

Cold-Start Nodes

New and small distributors have limited transaction history, causing GNNs to perform poorly on higher-risk entities where effective detection matters most.

OUR PROPOSED SOLUTION

Build a data ingestion layer with an entity resolution to merge fragmented records across systems into one unified transaction graph thus needing no new infrastructure.

Use inductive, temporal GNNs (GraphSAGE/TGN) with rolling retraining and drift detection to adapt to evolving supply chains, validated through synthetic seasonal benchmarks and historical backtesting

Since confirmed fraud examples are scarce, the model learns what normal trading looks like from abundant legitimate data, flagging any transaction that deviates from that pattern as suspicious, with no fraud labels required.

PRISM - THE AI RELIABILITY LAYER

WHAT WILL BE USING PRISM FOR?

Point to Note:

PRISM provides observability and evaluation around the AI system, however; the GNN remains the counterfeit-anomaly detection engine.

1

Monitor: Trace model inputs, outputs, latency & behaviour

2

Evaluate: Measure detection quality, failures & consistency

3

Detect: Identify recurring failures, drift & abnormal system behaviour

4

Improve: Convert reviewed failures into validated model/data improvements

In Overall:

GNN → PRISM → Human Review → Improvement

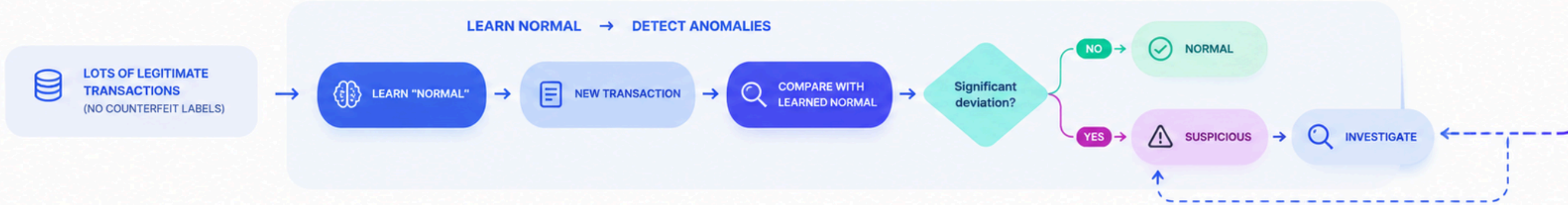
FROM ENTERPRISE DATA TO SAFER SUPPLY CHAINS

A complete pipeline — from your existing data to intelligent detection, monitoring and continuous improvement.



THE CORE IDEA

We learn what legitimate supply-chain behaviour looks like. Any transaction or entity that deviates from this pattern is flagged as suspicious — no counterfeit labels required.



IMPACT AND FUTURE SCOPE

Real World Impact:

Medicines:

Flag suspicious distributors, routes, and batches, helping investigators identify possible counterfeit injection points earlier.

Seeds & Fertilizers:

Account for seasonal demand while detecting unusual trading and distribution patterns.

Electronics:

Trace abnormal product movement, pricing, and distributor relationships across the supply network.

Key Benefits:

Label Efficient:

Learns from abundant legitimate transactions instead of relying on large counterfeit datasets.

Network-Aware

Understands relationships between companies, products, batches, and transactions.

Multi-Industry:

The same GNN framework can adapt to medicines, agriculture, electronics, and other supply chains.

Human-in-the-Loop:

Prioritizes suspicious cases for investigation instead of automatically declaring an entity guilty.

Future Scope:

Real Enterprise Data:

Integrate live enterprise data from invoices, logistics, and distribution systems.

Temporal GNNs:

Model how supply chain relationships and behaviors evolve over time.

Drift Detection:

Detect when changing or seasonal behavior makes the learned definition of “normal” outdated.

Real-Time Monitoring:

Continuously score transactions for faster detection and investigation.

Investigation AI / RAG:

Explain why an entity was flagged using relevant transactions and supporting evidence.